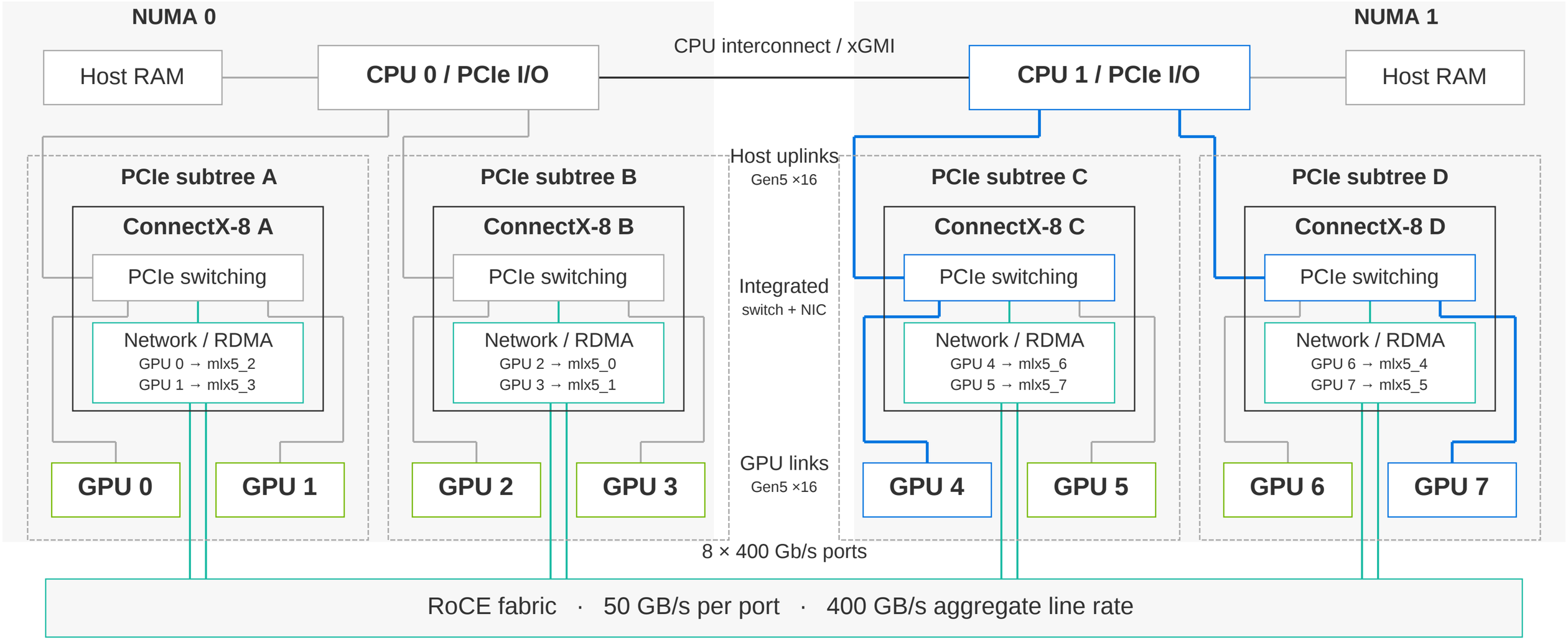
NUMA 0
NUMA 1
Host RAM
CPU 0 / PCIe I/O
CPU interconnect / xGMI
CPU 1 / PCIe I/O
Host RAM
Host uplinks
Gen5 ×16
PCIe subtree A
PCIe subtree B
PCIe subtree C
PCIe subtree D
ConnectX-8 A
ConnectX-8 B
ConnectX-8 C
ConnectX-8 D
PCIe switching
PCIe switching
PCIe switching
PCIe switching
Integrated
switch + NIC
Network / RDMA
GPU 0 → mlx5_2
GPU 1 → mlx5_3
Network / RDMA
GPU 2 → mlx5_0
GPU 3 → mlx5_1
Network / RDMA
GPU 4 → mlx5_6
GPU 5 → mlx5_7
Network / RDMA
GPU 6 → mlx5_4
GPU 7 → mlx5_5
GPU links
Gen5 ×16
GPU 0
GPU 1
GPU 2
GPU 3
GPU 4
GPU 5
GPU 6
GPU 7
8 × 400 Gb/s ports
RoCE fabric · 50 GB/s per port · 400 GB/s aggregate line rate